



URBAN REDEVELOPMENT AUTHORITY OF PITTSBURGH

INTERN PROJECT HANDOVER

LandCare Assurance

For this service period, which assigned parcels were serviced, by whom, and with what evidence.

[ura-gis/land-care-assurance](#)

[/monitoring/](#) · [/kpi/](#) · [/contractor/](#)

rutomo@ura.org (Rizaldy Utomo)

Contractor Compliance and Monitoring was lagging

THE TRIGGER

REPORTED

~11.7%

Blends Active and Request Only parcels, with no stated definition.

THE UNIVERSE

ASSIGNED

1,127

From a tag filter with **no ownership check** — sold parcels stayed in the denominator.

AT STAKE

ANNUAL SPEND

\$775k

Spend judged against evidence that **could not be traced to a parcel**.

WHY I BUILT IT THIS WAY

Outside GIS control: contractor compliance, Regrid's interface, field capacity.

Inside: the assignment query, the schemas, the metric definition, the published app.

A map gives leadership something to act on that a percentage cannot.

WHAT V1.0 DELIVERS

- **Map Monitor** — evidence displayed at the assignment polygon
- **KPI Dashboard** — completion, budget, contractor exposure by period
- **Contractor portal** — pick a parcel on the map or list, submit prefilled evidence
- **A stated metric** — period, denominator, source, and freshness on every number

One nightly job, everything else live at page load

Period: the 15th of one month through the 14th of the next

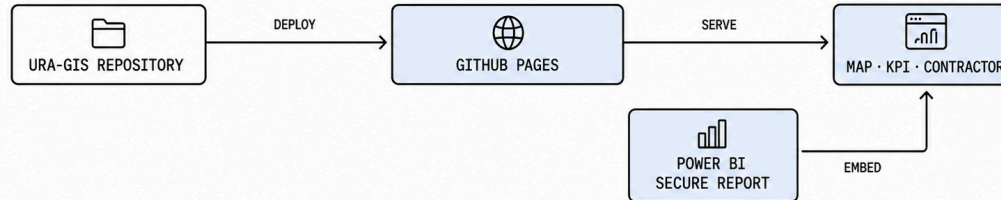
LandCare current data flow

— Live primary path

LIVE OPERATING PATH



PUBLIC APP DELIVERY



OWNERSHIP SPLIT

GIS and data operations own ingestion and store. Web and dashboard owner owns publish and application.

ADAPTER BOUNDARY

survey-layer.js and **assignment-layer.js** absorb every ArcGIS field rename. **Endpoints live nowhere else.**

DAILY CHECK

4:00 AM upstream ran · Map and KPI agree on the period count · Power BI embed loads · Pages green after code changes.

Power BI reports it, NetSuite records it

KPI dashboard · Land Care Budget tab
URA sign-in required

POWER BI

Authoritative reporting

Land Care Budget and Parcel Area are authenticated embeds, so each viewer sees the report through their own permissions and the numbers cannot drift from Finance.

Land Care Budget report · URA Finance workspace

NETSUITE

Upstream accounting

Saved search 1618 on account 66220 Lawn Maintenance. The importer in the repo is supervised reconciliation, not a feed.

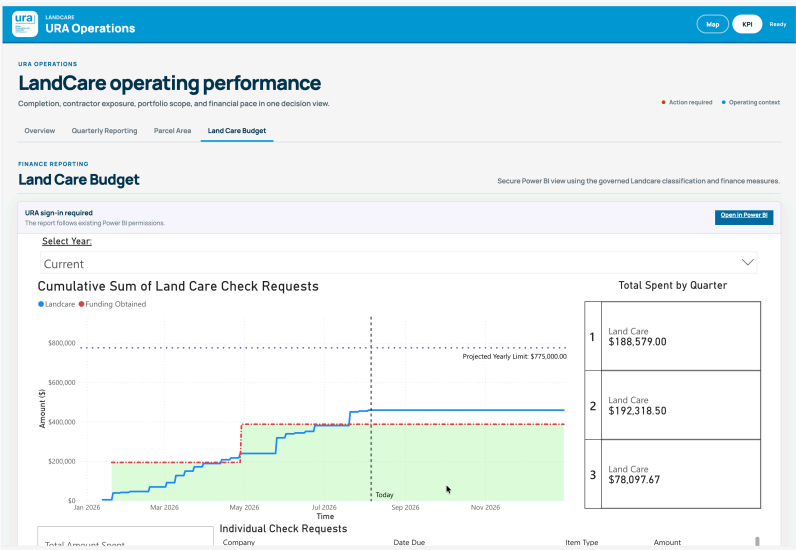
Imported on request, reconciled by hand

WORKBOOK

Contract expectations

Term, parcels, and forecast. The contracted shape of the work, held outside both systems.

Held by the contract manager, not in the repo



HOW THE EMBED WORKS

Land Care Budget and Parcel Area are secure Power BI embeds, not native cards. The page hosts the report; Microsoft authenticates each viewer, so the figures come straight from the Finance semantic model.

AUDITED 5 AUGUST 2026

SPENT TO DATE OF THE \$775,000 LIMIT

\$458,995.17 **59.23%**

Q1	Q2	Q3
\$188,579.00	\$192,318.50	\$78,097.67

Four blocks in every prompt

AGENTS.md is read automatically by Codex.
Keep it current.

A CONTEXT CALLING – FILES TO READ FIRST

```
Read AGENTS.md,  
handover/04-readiness-checklist.md,  
handover/01-owner-guide.md,  
docs/landcare-metrics-context.md first.
```

Inherits the source-of-truth and change rules before the agent plans anything.

B CONTRACT – GOAL, SOURCE, METRIC RULE

```
Goal: [outcome and which page it affects].  
Source: [ArcGIS item, table, published JSON].  
Metric rule: completion uses raw assignment-  
matched survey records over Active assigned;  
Request Only is excluded.
```

C TOOL CALLS – WHAT THE AGENT MUST RUN

```
python -m unittest discover -s tests  
node --test tests/survey-layer.test.mjs  
Pages validation + read-only ArcGIS smoke test
```

@chrome read the live route, its map and console

computer use drive the contractor flow, confirm the prefill

Name the tool. Report result, evidence, and rollback before commit.

D CUSTOM SNIPPETS – ADD PER TASK TYPE

Metric change: update the metrics doc and its test in the same change; show before and after for the latest period.

New page or restyle: use the design system rules; no framework, no build step.

Endpoint or field mapping: change only the adapter files; keep normalized property names stable.

Shared adapter: move every importer's ?v= version together.

SIX FAILURE MODES

Denominator drift

Hand-editing docs/landcare/data/

Stale cache-busting versions

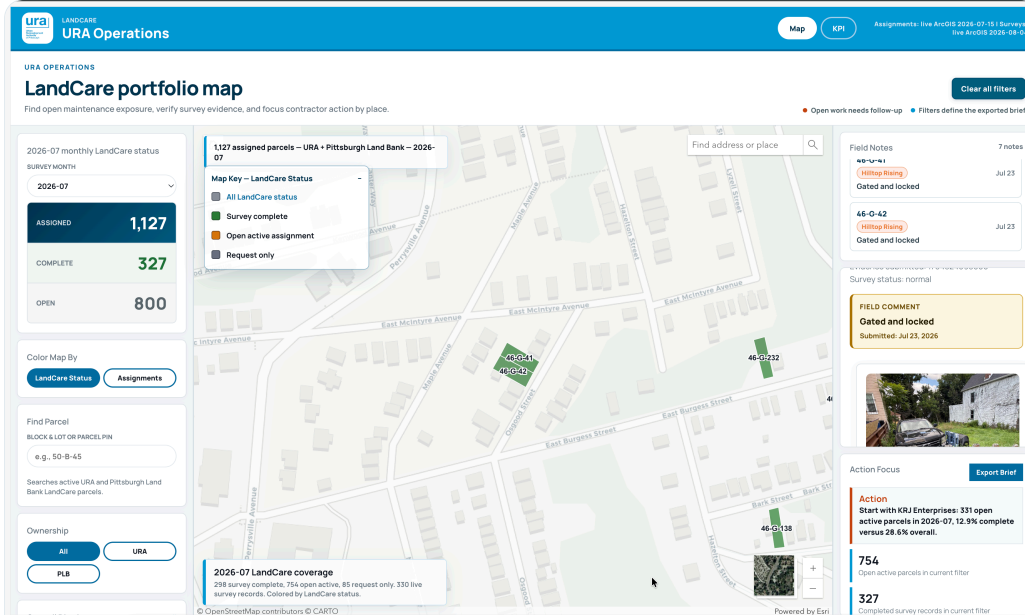
Guessing a NetSuite vendor alias

Hardcoded endpoints outside the adapters

Calling the native finance cards Power BI-backed

Reusable Style for future GIS Web App

Manrope only · four hues
plus ink, muted, and line neutrals



Map Monitor, live · fixed blue header, pill filters, hairline cards, and status colours always paired with a label.

APP.CSS

~2,800 lines, coupled to the ArcGIS theme. Edit to change an existing page.

EXECUTIVE-BI.CSS

156 portable lines. Copy to start a new dashboard, with design-system/example.html.

TOKENS OVER COMPONENTS

- -bi-primary #006c9f · analysis, active controls
- -bi-deep #00334f · headings, authoritative detail
- -bi-risk #c2410c · action, risk, follow-up
- -bi-success #2e7d32 · target, complete, stable

Override the variables at .bi-dashboard level rather than restyling a component.

FIXED PAGE ORDER

- Masthead
- Tabs
- Context + filters
- Decision question
- 3-5 metrics
- Dominant map or chart
- Operational ledger

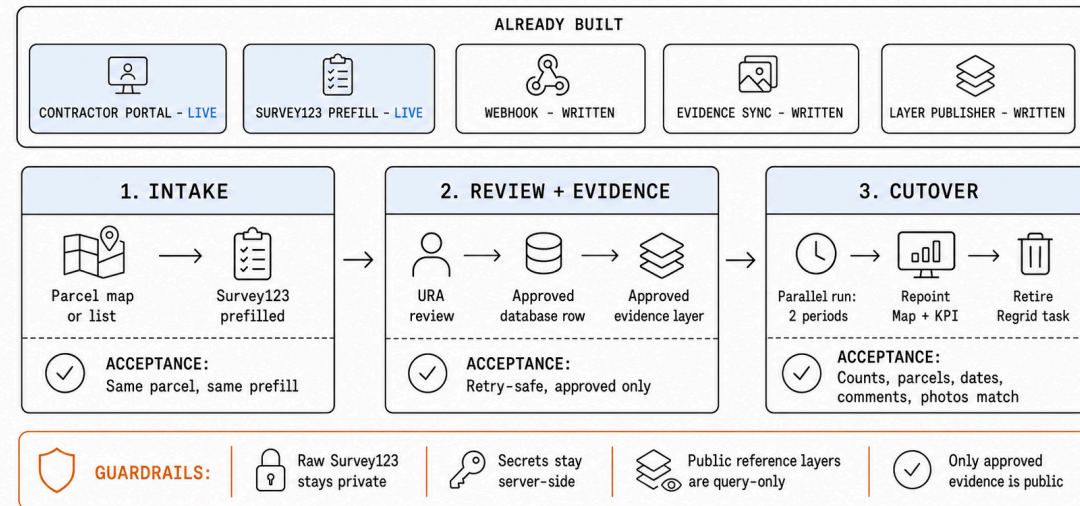
Checked in review: colour always paired with a label; every number shows source, period, denominator, freshness; contrast $\geq 4.5:1$; explicit loading, empty, stale, error states.

The scope is one intake form and one export

Assignments, parcel geometry, ownership, dashboard and portal are already URA property

Replace Regrid: an intake cutover

URA already owns the platform



STILL GROUNDED IN ARCGIS

Parcels, assignments, and evidence stay **hosted feature layers**, queried live at page load. URA owns the intake; Esri carries the geometry and sharing model.

STILL IN DEVELOPMENT

Submitted surveys pipeline need to be checked. Webhook, evidence sync, and layer publisher are written but not deployed, to fully replace regrid. **Deploying that path is the last step before cutover.**

Keep the source, the metric, and the release path stable

Status: [handover/04-readiness-checklist.md](#)
Deep reference: [handover/01-owner-guide.md](#)

1 · DAILY CHECK

4 AM ArcGIS
freshness



Map and KPI agree



Power BI embed
loads

2 · SAFE CHANGE

Read
sources



Define
metric



Edit
adapter



Run tests



PR +
deploy

3 · DO NOT DRIFT

- ! Raw matched records, not unique parcels
- ! Move shared ?v= versions together
- ! Endpoints live in the adapter files
- ! Never hand-edit docs/landcare/data/
- ! Never embed Power BI page 8c93bab49c96aa8e3bd2 — wrong scope

4 · DESIGN SYSTEM

Existing product
docs/landcare/app.css

New dashboard
docs/design-system/executive-
bi.css

Decision first · 3 to 5 KPIs · one dominant map or chart · operational ledger

OPEN ITEMS FOR THE NEXT OWNER TO DECIDE

Survey123 evidence path is code-complete but not deployed. The single blocker on replacing Regrid.

Power BI workspace administration is not yet granted. Refresh history and service-principal access stay blocked until it is.

The deprecated 7 AM task still needs archiving and disabling on the VM — not deleting on day one.